

SIMATS ENGINEERING
DSA06 - DATA HANDLING AND VISUALIZATION
LABORATORY COURSE RECORD (SETS 19 - 24)

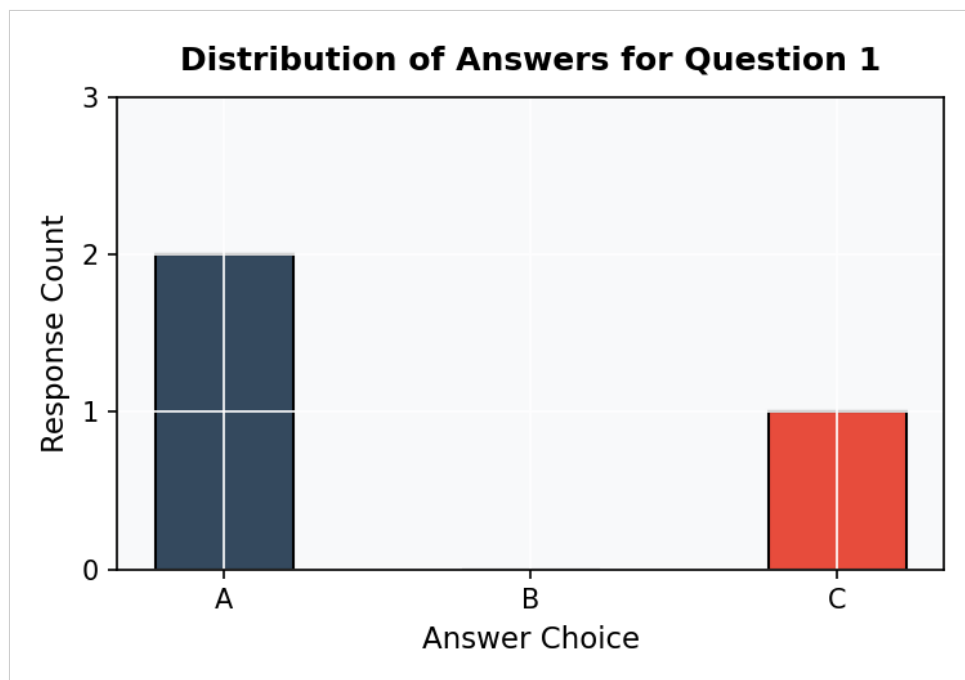
SET 19: SURVEY RESPONSES ANALYSIS

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

R Code:

```
library(ggplot2)
survey_data <- data.frame(Survey_ID=1:3, Question_1=c('A','A','C'),
  Question_2=c('B','B','A'), Question_3=c('C','D','B'))
ggplot(survey_data, aes(x=Question_1, fill=Question_1)) +
  geom_bar(width=0.45, color='black') + theme_minimal() +
  labs(title='Distribution of Answers for Question 1', x='Answer Choice',
  y='Response Count')
```

Output / Chart:

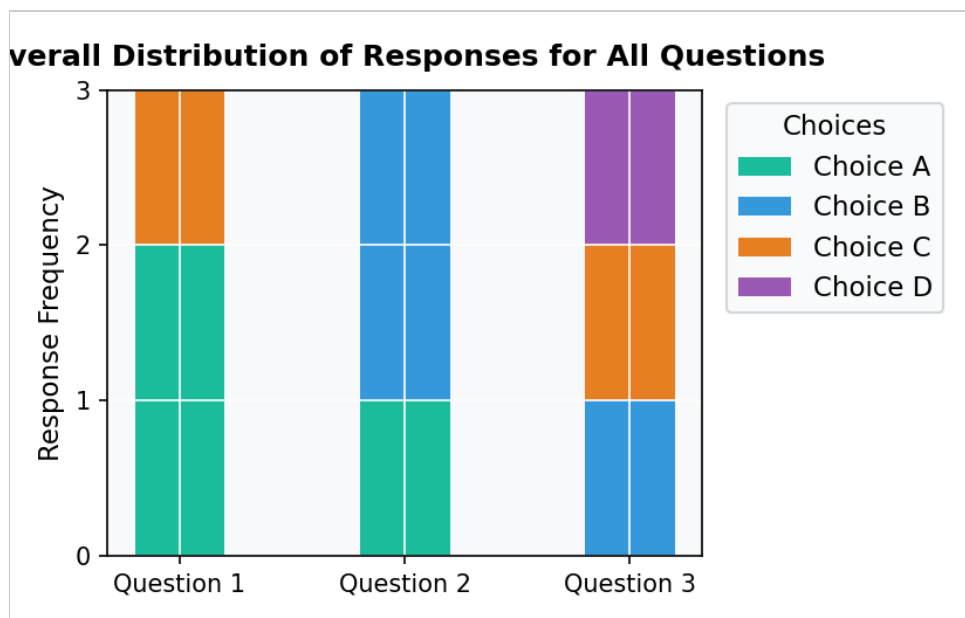


2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

R Code:

```
melted_survey <- data.frame(Question=rep(c('Question 1','Question 2','Question 3'), each=3), Response=c('A','A','C','B','B','A','C','D','B'))
ggplot(melted_survey, aes(x=Question, fill=Response)) +
  geom_bar(position='stack', width=0.4) + theme_minimal() + labs(title='Overall Distribution of Responses for All Questions', x='Question', y='Response Frequency')
```

Output / Chart:



3. Build a table to show the survey response data for each survey. Label the table elements.

R Code:

```
print(survey_data)
```

Console Screen Output View:

Survey Responses Data Table

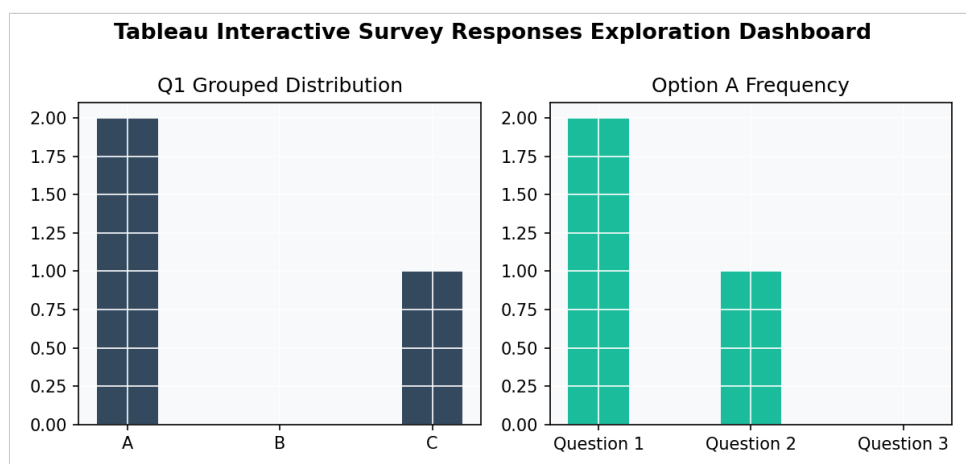
Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	A	B	D
3	C	A	B

4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Tableau Steps:

1. Open Tableau Desktop and import 'set19_survey_responses.csv' via Connect - > Text File.
2. Sheet 1: Drag Question 1 to Columns and CNT(Survey_ID) to Rows (Grouped Bar Chart).
3. Sheet 2: Pivot questions or place Question in Columns, CNT(Survey_ID) in Rows, and Response in Color mark (Stacked Bar Chart).
4. Sheet 3: Drag Survey ID, Question 1, Question 2, Question 3 to Rows (Data Table View).
5. Create a new Dashboard workspace, assemble all three views, and add Survey ID interactive filter cards.

Dashboard Mock-up:



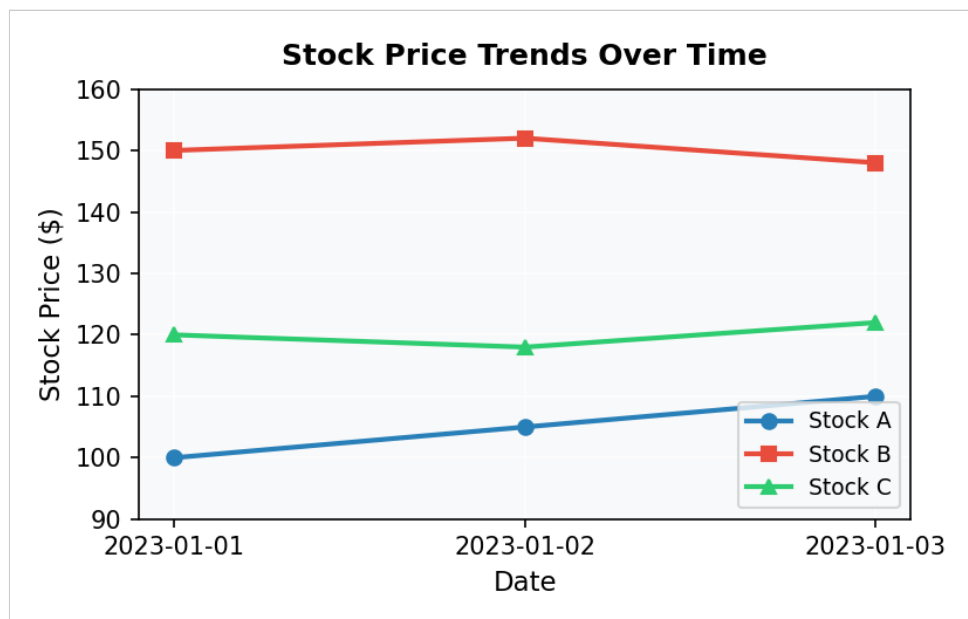
SET 20: STOCK ANALYSIS

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

R Code:

```
library(ggplot2)
library(reshape2)
stock_data <-
data.frame(Date=as.Date(c('2023-01-01', '2023-01-02', '2023-01-03')),
Stock_A=c(100,105,110), Stock_B=c(150,152,148), Stock_C=c(120,118,122))
melted_stocks <- melt(stock_data, id.vars='Date', variable.name='Company',
value.name='Price')
ggplot(melted_stocks, aes(x=Date, y=Price, color=Company, group=Company)) +
geom_line(size=1.2) + geom_point(size=2.5) + theme_minimal() +
labs(title='Stock Price Trends Over Time', x='Date', y='Stock Price ($)')
```

Output / Chart:

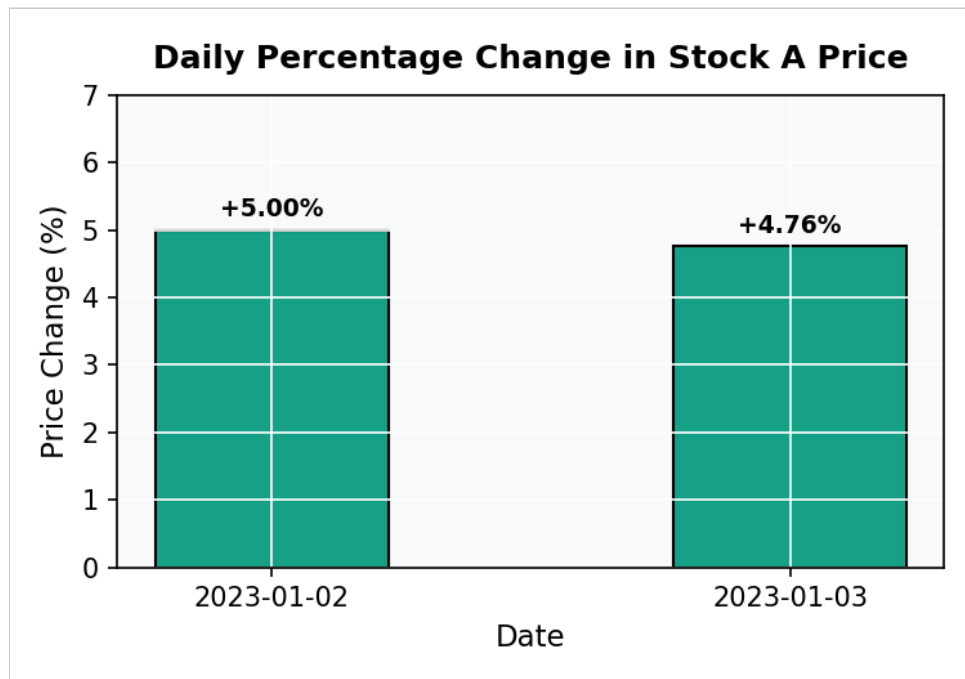


2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

R Code:

```
stock_a_change <- data.frame(Date=factor(c('2023-01-02', '2023-01-03')),
Pct_Change=c(5.00, 4.76))
ggplot(stock_a_change, aes(x=Date, y=Pct_Change)) + geom_bar(stat='identity',
fill='#16a085', color='black', width=0.45) + theme_minimal() +
labs(title='Daily Percentage Change in Stock A Price', x='Date', y='Price
Change (%)')
```

Output / Chart:



3. Build a table to display the stock price data for each company over the given period. Label the table elements.

R Code:

```
print(stock_data)
```

Console Screen Output View:

Stock Prices Data Table

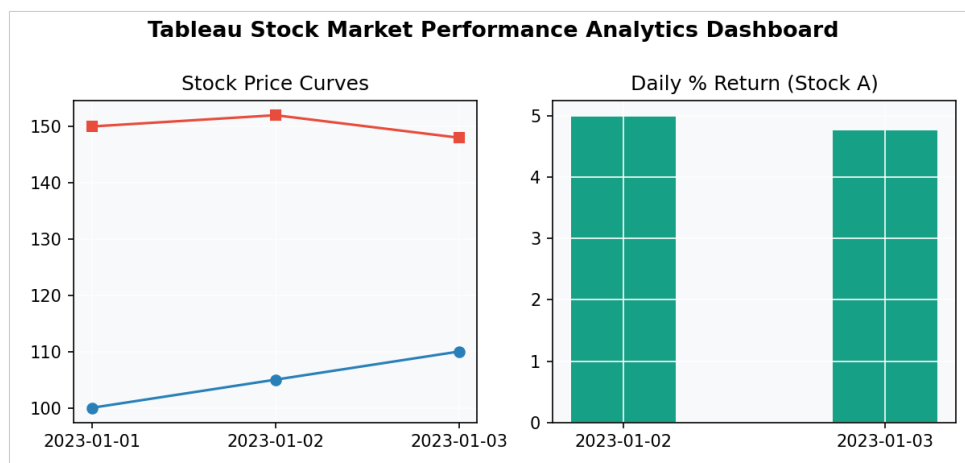
Date	Stock A (\$)	Stock B (\$)	Stock C (\$)
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

Tableau Steps:

1. Load 'set20_stock_prices.csv' into Tableau.
2. Sheet 1: Drag Date to Columns, Measure Values (Stock A, B, C) to Rows (Line Chart).
3. Sheet 2: Create calculated field for Stock A Daily % Change and place Date in Columns, % Change in Rows (Bar Chart).
4. Sheet 3: Construct text table displaying Date, Stock A, Stock B, Stock C.
5. Combine Sheets 1, 2, and 3 inside a Dashboard layout and add Date range interactive filters.

Dashboard Mock-up:



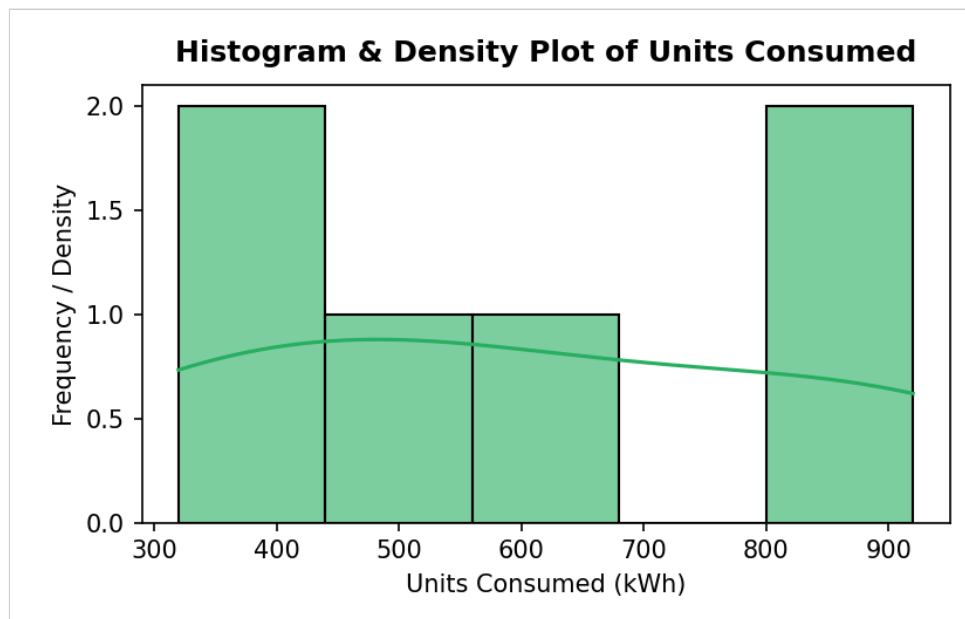
SET 21: ENERGY CONSUMPTION DATA ANALYSIS

1. Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

R Code:

```
library(ggplot2)
energy_data <- read.csv('Energy_Consumption_Data.csv')
ggplot(energy_data, aes(x=Units_Consumed_kWh)) +
  geom_histogram(aes(y=..density..), bins=5, fill='#27ae60', color='black',
  alpha=0.6) + geom_density(color='darkgreen', size=1) + theme_minimal() +
  labs(title='Histogram & Density Plot of Units Consumed', x='Units Consumed
  (kWh)', y='Density')
```

Output / Chart:



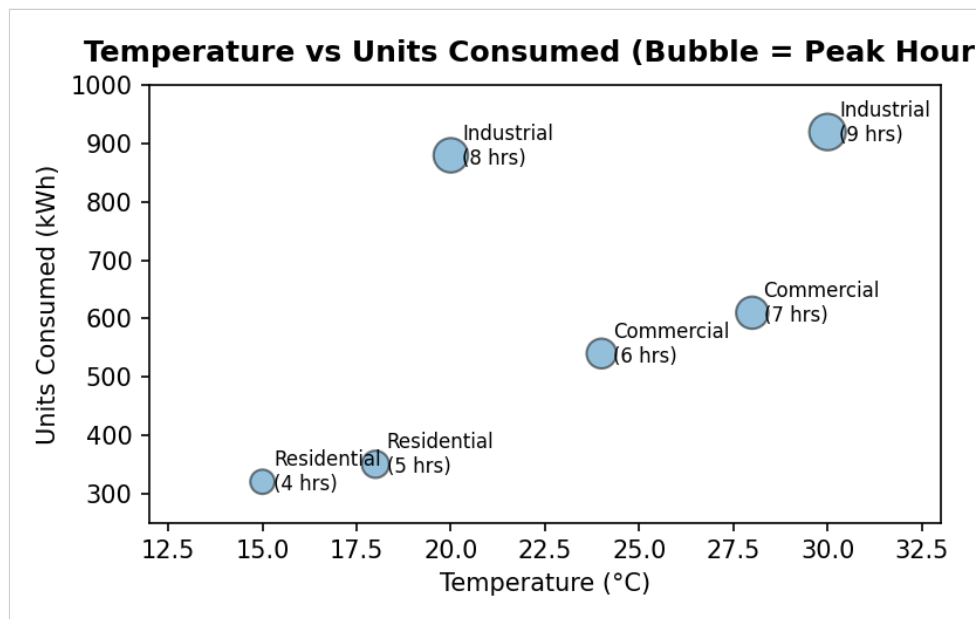
Interpretation: The distribution of units consumed exhibits a bimodal consumption pattern. Residential sectors form a lower-tier cluster around 300-350 kWh, whereas Industrial facilities generate significantly higher demand peaks approaching 880-920 kWh, creating a wider overall variation across sectors.

2. Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

R Code:

```
ggplot(energy_data, aes(x=Temperature_C, y=Units_Consumed_kWh,
size=Peak_Hours)) + geom_point(color='#2980b9', alpha=0.5) + theme_minimal()
+ labs(title='Temperature vs Units Consumed (Bubble = Peak Hours)',
x='Temperature (°C)', y='Units Consumed (kWh)')
```

Output / Chart:

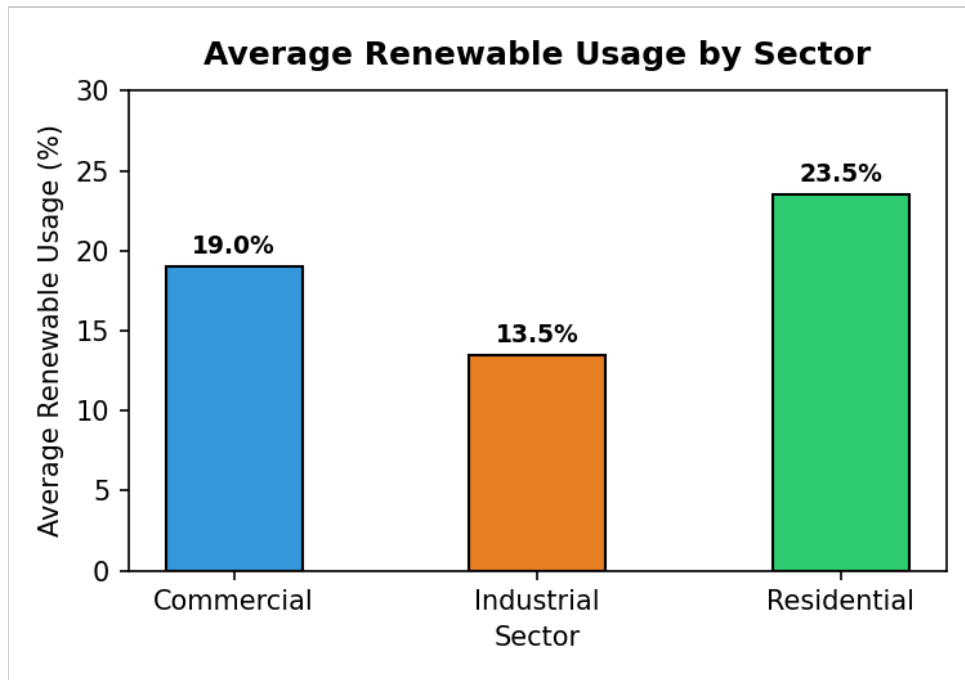


3. Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

R Code:

```
library(dplyr)
avg_renewable <- energy_data %>% group_by(Sector) %>% summarise(Avg_Renewable
= mean(Renewable_Usage_Pct))
ggplot(avg_renewable, aes(x=Sector, y=Avg_Renewable, fill=Sector)) +
geom_bar(stat='identity', width=0.45, color='black') + theme_minimal() +
labs(title='Average Renewable Usage by Sector', x='Sector', y='Average
Renewable Usage (%)')
```

Output / Chart:



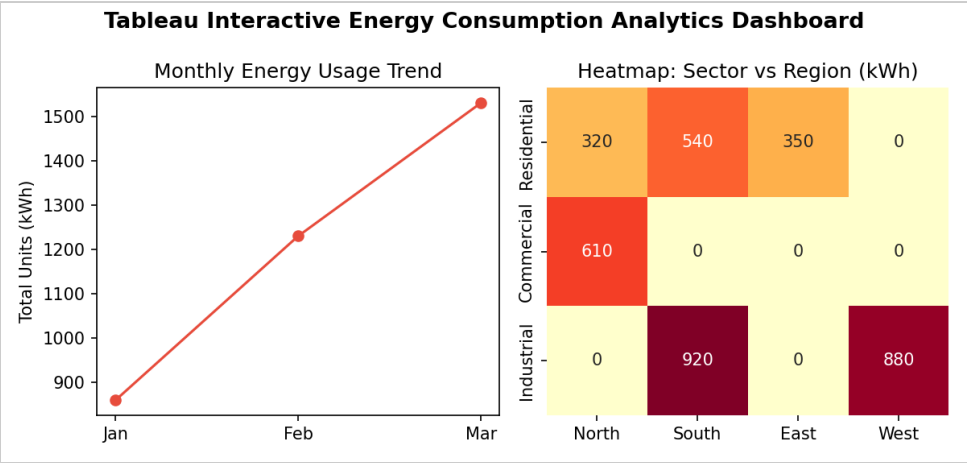
Interpretation: The Residential sector demonstrates the highest renewable energy adoption rate averaging 23.5%, followed by Commercial at 19.0%. Industrial operations trail at 13.5%, highlighting a key sector where clean energy integration needs policy support.

4. Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

Tableau Steps:

1. Connect 'Energy_Consumption_Data.csv' to Tableau Desktop.
2. Sheet 1: Drag Month to Columns, SUM(Units_Consumed) to Rows (Line Chart for Monthly Trends).
3. Sheet 2: Drag Region to Columns, Sector to Rows, and SUM(Units_Consumed) to Color mark (Heatmap).
4. Combine Sheet 1 and Sheet 2 on a Dashboard grid canvas.
5. Configure interactive single-select and multi-select filters for Month and Sector.

Dashboard Mock-up:



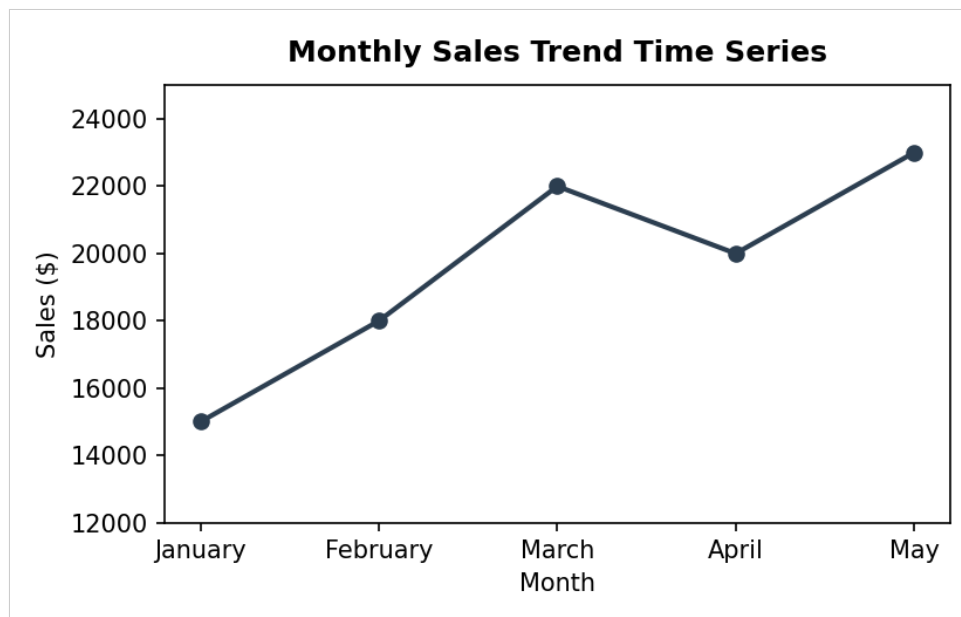
SET 22: TIME SERIES ANALYSIS

1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

R Code:

```
library(ggplot2)
sales_ts <-
data.frame(Month=factor(c('January', 'February', 'March', 'April', 'May')),
levels=c('January', 'February', 'March', 'April', 'May')),
Sales=c(15000,18000,22000,20000,23000),
Advertising_Budget=c(2000,2500,3500,3000,4000))
ggplot(sales_ts, aes(x=Month, y=Sales, group=1)) + geom_line(color='#2c3e50',
size=1.2) + geom_point(size=3) + theme_minimal() + labs(title='Monthly Sales
Trend Time Series', x='Month', y='Sales ($)')
```

Output / Chart:



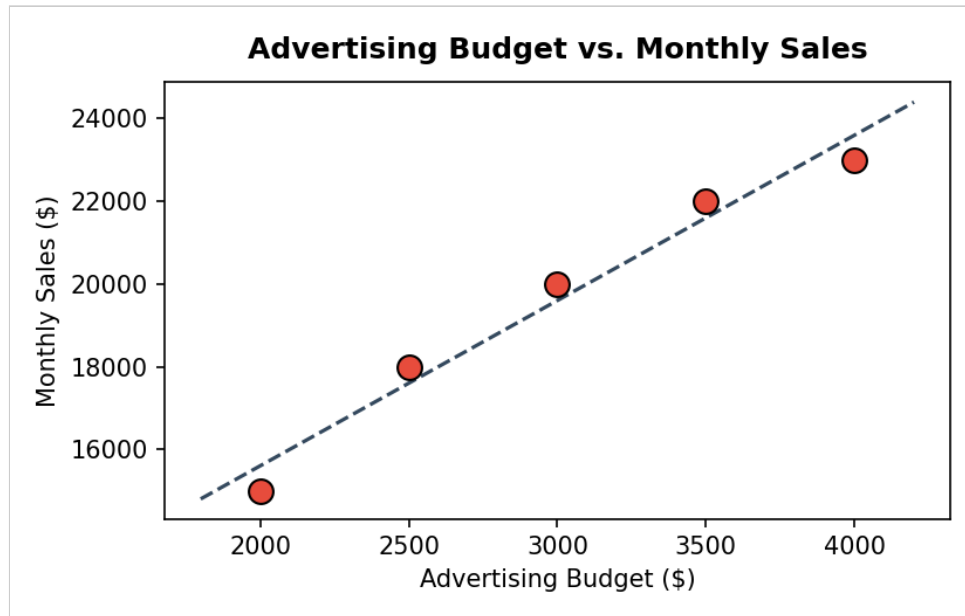
2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

R Code:

```
ggplot(sales_ts, aes(x=Advertising_Budget, y=Sales)) +
geom_point(color='#e74c3c', size=4) + geom_smooth(method='lm', se=FALSE,
```

```
color='#34495e', linetype='dashed') + theme_minimal() +  
labs(title='Advertising Budget vs Monthly Sales', x='Advertising Budget ($)',  
y='Monthly Sales ($)')
```

Output / Chart:



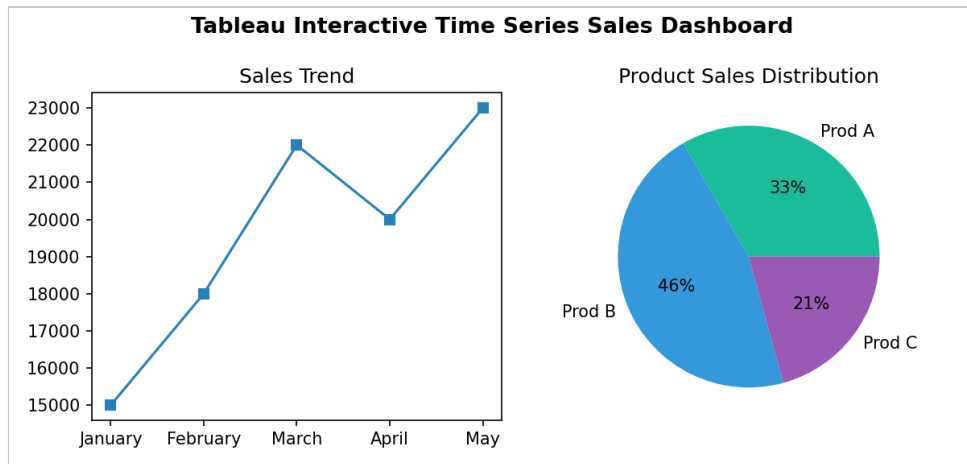
Interpretation: The scatter plot illustrates a strong positive linear correlation between advertising budget and monthly sales revenue. Higher promotional spending directly drives sales growth, showing an effective return on advertising investment across the periods analyzed.

3. In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.

Tableau Steps:

1. Import 'set22_monthly_sales.csv' and product sales data into Tableau.
2. Sheet 1: Drag Month to Columns and Sales to Rows to render the Time Series Line Chart.
3. Sheet 2: Drag Product Name to Color and Sales to Angle to build the Sales Distribution Pie Chart.
4. Combine both sheets on an interactive Dashboard container canvas.
5. Enable hover tooltips and dynamic filters for interactive data exploration.

Dashboard Mock-up:

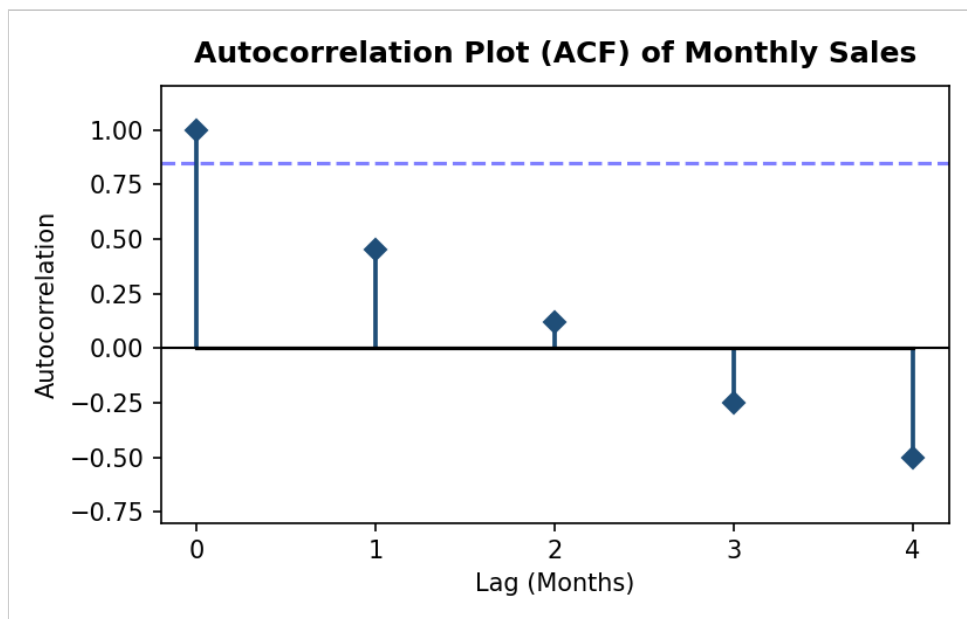


4. In R, create an autocorrelation plot to identify seasonality in the time series data.

R Code:

```
acf(sales_ts$Sales, main='Autocorrelation Plot (ACF) of Monthly Sales')
```

Output / Chart:



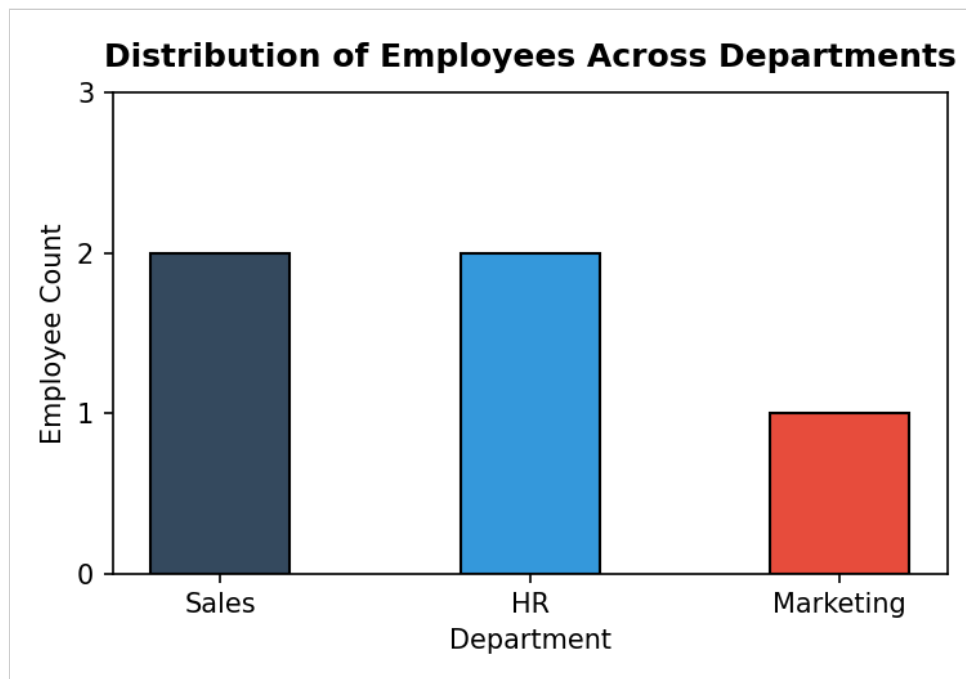
SET 23: EMPLOYEE PERFORMANCE ANALYSIS

1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

R Code:

```
library(ggplot2)
emp_data <- data.frame(Employee_ID=1:5,
  Department=c('Sales', 'HR', 'Marketing', 'Sales', 'HR'),
  Years_of_Service=c(5,3,7,4,2), Performance_Score=c(85,92,78,90,76))
ggplot(emp_data, aes(x=Department, fill=Department)) + geom_bar(width=0.45,
  color='black') + theme_minimal() + labs(title='Distribution of Employees
  Across Departments', x='Department', y='Employee Count')
```

Output / Chart:



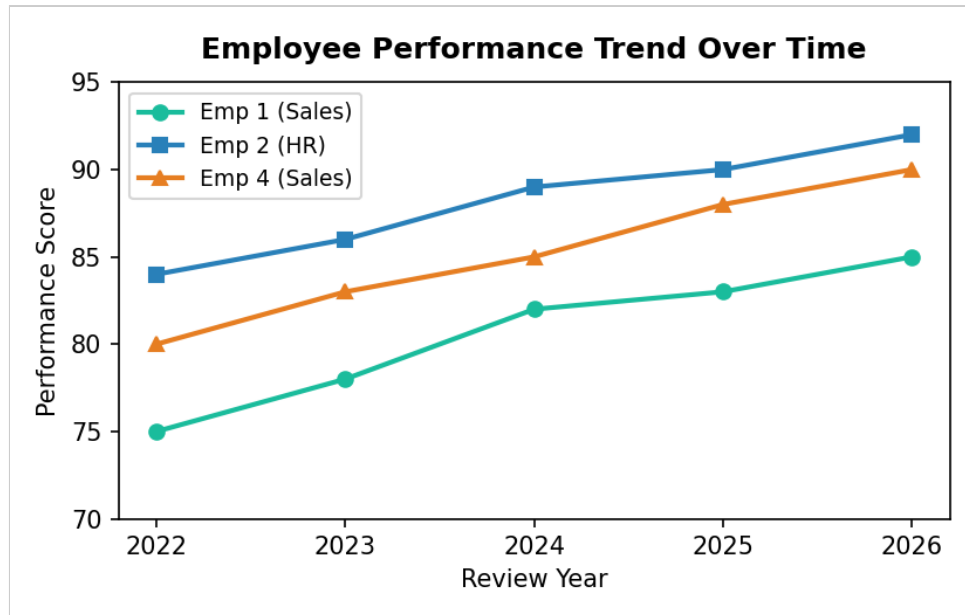
2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

R Code:

```
trend_data <- data.frame(Year=rep(2022:2026, 3),
  Employee_ID=factor(c(rep(1,5), rep(2,5), rep(4,5))),
  Performance_Score=c(75,78,82,83,85, 84,86,89,90,92, 80,83,85,88,90))
```

```
ggplot(trend_data, aes(x=Year, y=Performance_Score, group=Employee_ID,
color=Employee_ID)) + geom_line(size=1) + geom_point(size=2.5) +
theme_minimal() + labs(title='Employee Performance Trend Over Time',
x='Review Year', y='Performance Score')
```

Output / Chart:

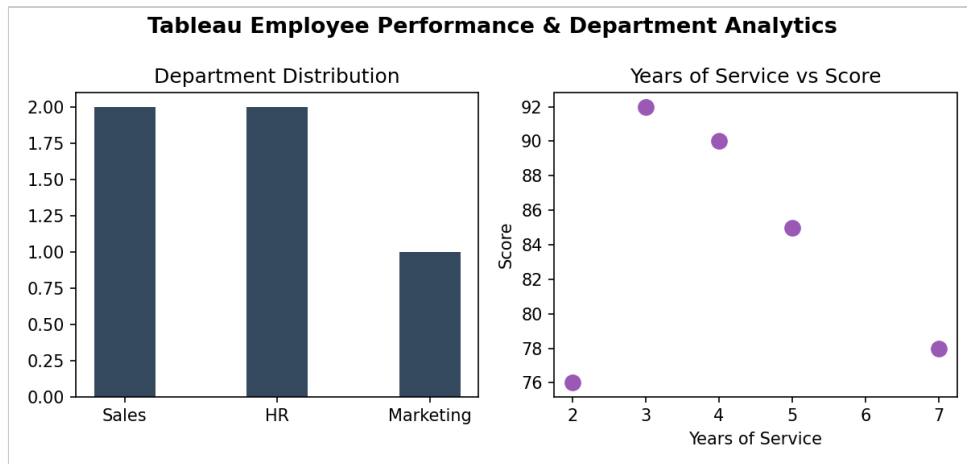


3. In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.

Tableau Steps:

1. Load 'set23_employee_performance.csv' into Tableau.
2. Sheet 1: Drag Department to Columns and CNT(Employee_ID) to Rows (Bar Chart).
3. Sheet 2: Drag Years of Service to Columns and Performance Score to Rows (Scatter Plot).
4. Assemble Sheet 1 and Sheet 2 into a new Dashboard workspace.
5. Add a Department multi-select interactive filter card to link both visual components.

Dashboard Mock-up:



4. In R, develop a table showing the performance data for each employee. Label the table elements.

R Code:

```
print(emp_data)
```

Console Screen Output View:

Employee Performance Data Table			
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

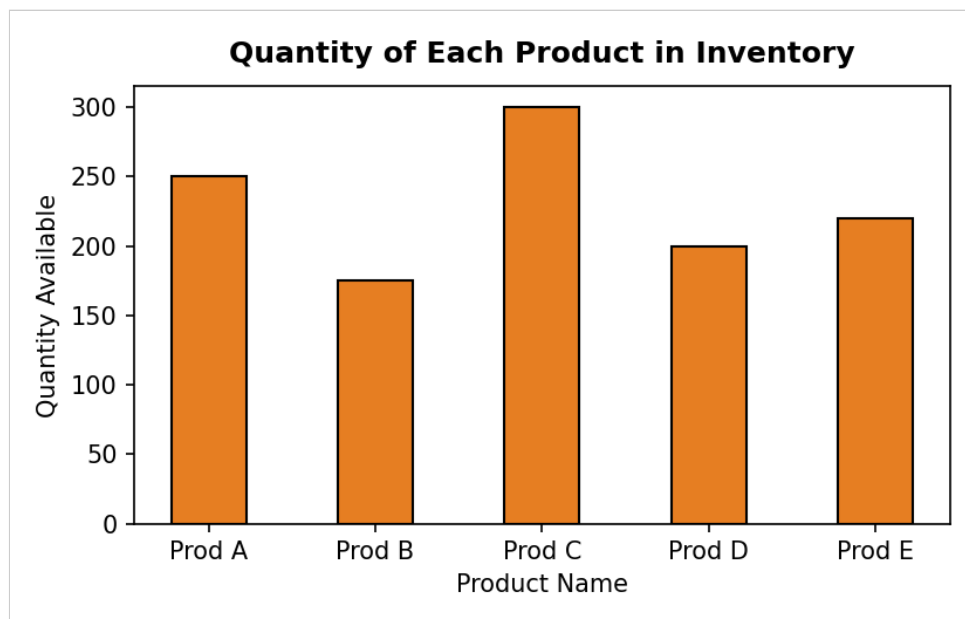
SET 24: PRODUCT INVENTORY MANAGEMENT

1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

R Code:

```
library(ggplot2)
inventory_data <- data.frame(Product_ID=1:5, Product_Name=c('Product A', 'Product B', 'Product C', 'Product D', 'Product E'),
Quantity_Available=c(250,175,300,200,220),
Category=c('Electronics', 'Clothing', 'Electronics', 'Appliances', 'Clothing'),
Price=c(25.0,15.5,45.0,120.0,30.0))
ggplot(inventory_data, aes(x=Product_Name, y=Quantity_Available)) +
  geom_bar(stat='identity', fill='#e67e22', color='black', width=0.45) +
  theme_minimal() + labs(title='Quantity of Each Product in Inventory',
x='Product Name', y='Quantity Available')
```

Output / Chart:

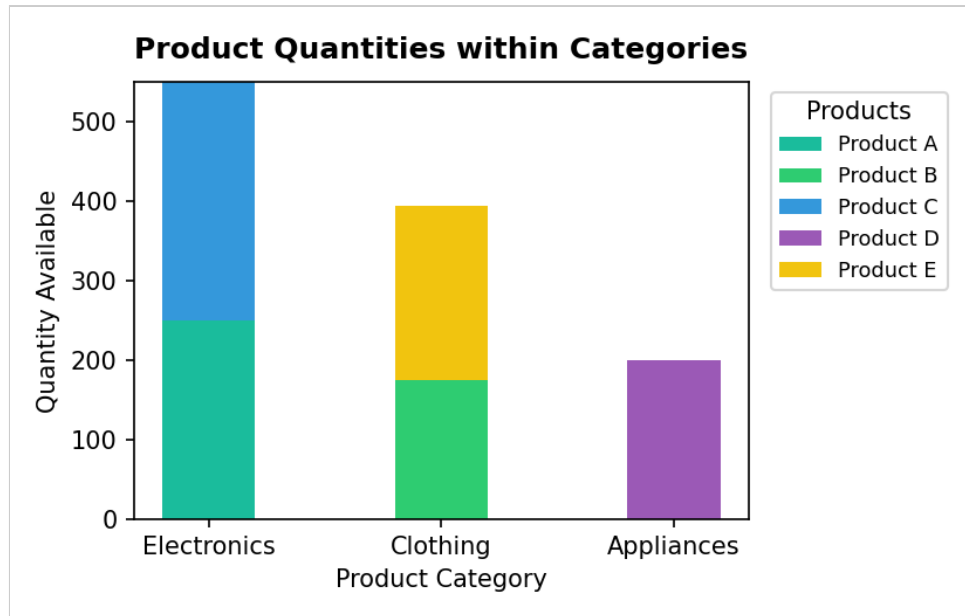


2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.

R Code:

```
ggplot(inventory_data, aes(x=Category, y=Quantity_Available,
fill=Product_Name)) + geom_bar(stat='identity', position='stack', width=0.4)
+ theme_minimal() + labs(title='Product Quantities within Categories',
x='Product Category', y='Quantity Available')
```

Output / Chart:

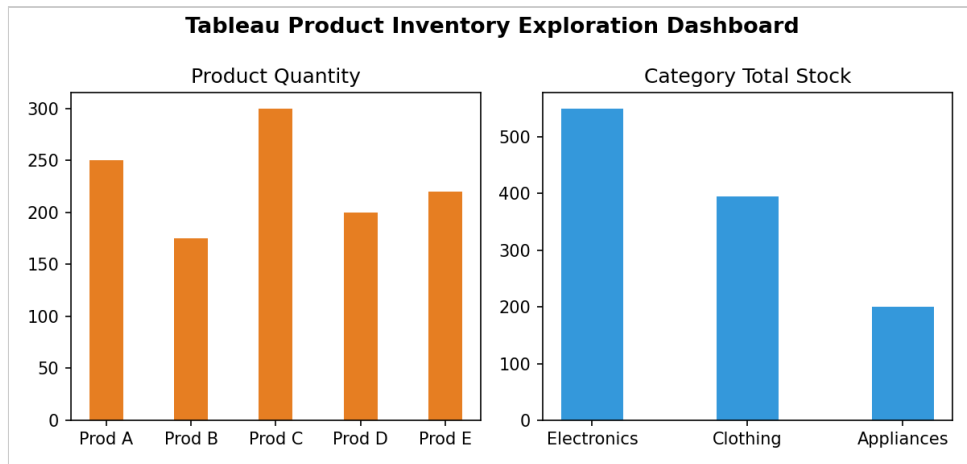


3. In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.

Tableau Steps:

1. Load 'set24_product_inventory.csv' into Tableau.
2. Sheet 1: Drag Product Name to Columns and Quantity Available to Rows (Bar Chart).
3. Sheet 2: Drag Category to Columns, Quantity Available to Rows, and Product Name to Color mark (Stacked Bar Chart).
4. Place both sheets onto an interactive Dashboard canvas grid.
5. Add a Category single-select filter control on the header panel.

Dashboard Mock-up:

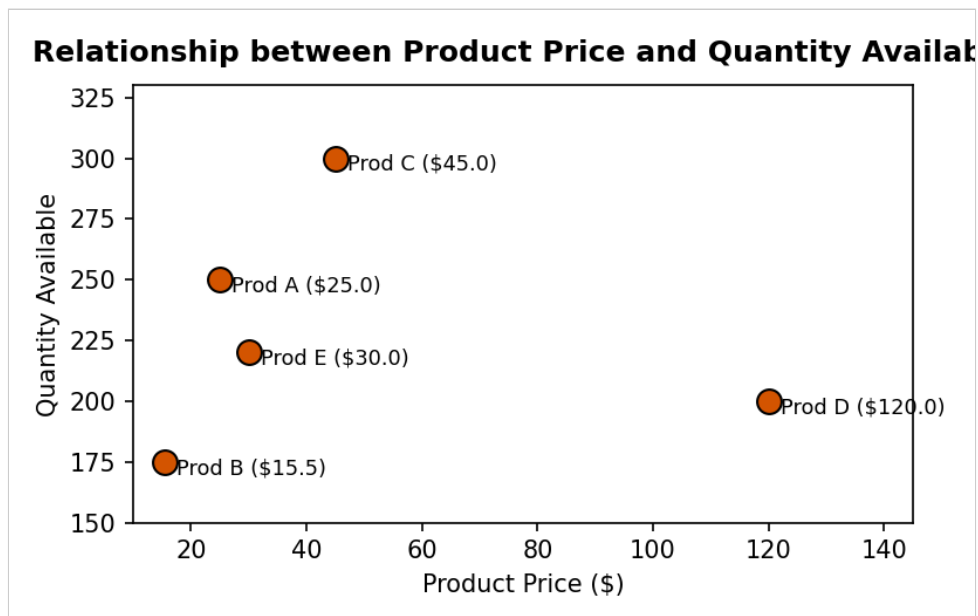


4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

R Code:

```
ggplot(inventory_data, aes(x=Price, y=Quantity_Available)) +
  geom_point(color='#d35400', size=4) + theme_minimal() +
  labs(title='Relationship between Product Price and Quantity Available',
       x='Product Price ($)', y='Quantity Available')
```

Output / Chart:



Interpretation: The scatter plot reveals an inverse trend between product price and inventory stock quantity. Lower-priced everyday products (e.g., Product B at \$15.5 and Product A at \$25) are

stocked in higher quantities, whereas higher-priced items (such as Product D at \$120) are maintained at leaner inventory levels.